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Department-Artificial Intelligence and Data Science

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Lab Practical

Data Visualization from Extraction Transformation and Loading (ETL) Process.

Step 1: Check Data (Extraction)

```
import pandas as pd
import matplotlib.pyplot as plt

# EXTRACT - Load raw data from CSV
df = pd.read_csv("Superstore.csv", encoding='latin1')

print("Shape of data:", df.shape)
print("\nFirst 5 rows:")
df.head()
```

✓ 5.2s

Shape of data: (9994, 21)

Step 2: Transform

Check-missing values

```
# Check missing values
print("Missing values:\n", df.isnull().sum())

# Drop rows with null values
df = df.dropna()

# Select only required columns
df = df[['Category', 'Sub-Category', 'Sales', 'Profit', 'Quantity']]

✓ 0.0s
```

Missing values:

Row ID	0
Order ID	0
Order Date	0
Ship Date	0
Ship Mode	0
Customer ID	0
Customer Name	0
Segment	0
Country	0
City	0
State	0
Postal Code	0
Region	0
Product ID	0
Category	0
Sub-Category	0
Product Name	0
Sales	0
Quantity	0
Discount	0
Profit	0

dtype: int64

Remove Duplicates if any

```
df = df.drop_duplicates()

print("\nCleaned Data Shape:", df.shape)
print(df.head())
```

✓ 0.3s

Cleaned Data Shape: (7729, 5)

	Category	Sub-Category	Sales	Profit	Quantity
0	Furniture	Bookcases	261.9600	41.9136	2
1	Furniture	Chairs	731.9400	219.5820	3
2	Office Supplies	Labels	14.6200	6.8714	2
3	Furniture	Tables	957.5775	-383.0310	5
4	Office Supplies	Storage	22.3680	2.5164	2

Step 3: Load

```
# LOAD - Save cleaned data to new CSV
df.to_csv("Superstore_cleaned.csv", index=False)

print("Cleaned data loaded/saved successfully!")
```

✓ 0.0s

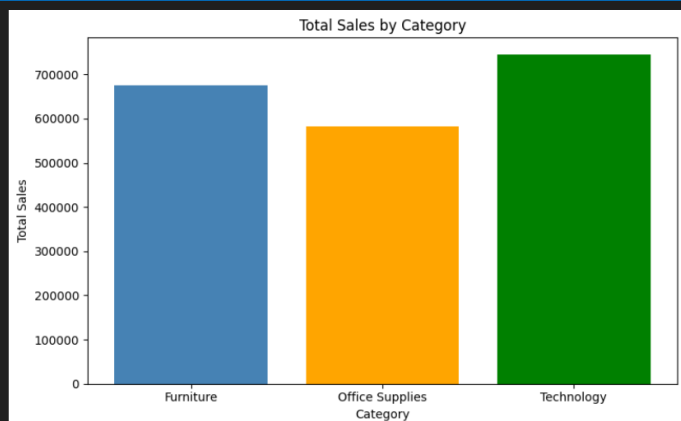
Cleaned data loaded/saved successfully!

Step 4: Visualization

```
category_sales = df.groupby('Category')['Sales'].sum()

plt.figure(figsize=(8,5))
plt.bar(category_sales.index, category_sales.values, color=['steelblue', 'orange', 'green'])
plt.title('Total Sales by Category')
plt.xlabel('Category')
plt.ylabel('Total Sales')
plt.tight_layout()
plt.show()
```

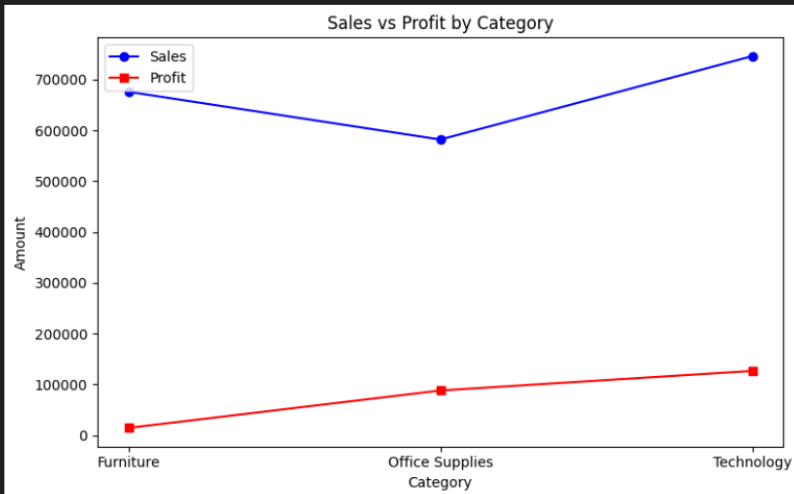
✓ 0.3s



```
category_profit = df.groupby('Category')['Profit'].sum()

plt.figure(figsize=(8,5))
plt.plot(category_sales.index, category_sales.values, marker='o', label='Sales', color='blue')
plt.plot(category_profit.index, category_profit.values, marker='s', label='Profit', color='red')
plt.title('Sales vs Profit by Category')
plt.xlabel('Category')
plt.ylabel('Amount')
plt.legend()
plt.tight_layout()
plt.show()
```

✓ 1.0s



Sign: _____.